MILITARY SPECIFICATION

MANUAL, TECHNICAL: NONDESTRUCTIVE INSPECTION

1. SCOPE

1.1 This specification contains instructions for the preparation of technical manuals required for the application of nondestructive inspection (NDI) methods on aircraft (-36) manual), missiles and space weapons (-26manuals), and engines (-9 manual). Instructions for the preparation of NDI methods to be included in maintenance manuals for components and equipment not requiring a separate manual (-36, -26, -9 are also provided.

1.2 The manuals and procedures covered by this specification are intended to provide instructions and guidance to NDI technicians in the organizational, intermediate, and depot levels of maintenance for the use of NDI methods on systems, subsystems, support systems and their components.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents, of the issue in effect on the date of invitation for bids, request proposal, or contract date, whichever occurs first, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MIL-M-38784 Manuals, Technical; General Requirements for for Preparation of Printing
MIL-P-38790 Printing Production of Technical Manuals: General Requirements for

MILITARY STANDARDS

MIL-STD-882 System Safety Program for Systems and Associated Subsystems, and Equipment, Requirements for

PUBLICATIONS

Technical Orders

00-5-1 Air Force Technical Order System

00-20-5 Aircraft, Drone and Air Launched Missile Flight Reports and Supporting Maintenance Records

33B-1-1 Nondestructive Inspection Methods

Table of Allowances

TA-455 Table of Allowances for Nondestructive Inspection Equipment

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the latest issue in effect on the dateof invitation for bids, or request for proposal shall apply.

Nondestructive Testing Handbook, Robert C. McMaster,ed.

(Application for copies should be addressed to the American Society for Nondestructive Testing, Inc., 914 Chicago Avenue, Evanston, Illinois 6202)

3. REQUIREMENTS

3.1 Preparation. Unless otherwise specified, the general manner of preparation of the NDI manual shall be in accordance with MIL-M-38784 and MIL-STD-882.

3.1.1 Development of a NDI manual shall meet the individual needs of the system for which it is intended. Examples of system design concepts with differences that must be considered are: fighter vs cargo vs helicopter aircraft; and jet vs reciprocating engines.

3.1.2 NDI procedures for components or equipment that will not be extensive enough to justify a separate manual, shall be placed in the inspection section of the applicable manual. Examples of components or equipment that may not require a separate NDI manual are: aerospace ground equipment; accessory gear drives; electrical generators; auxiliary power units, etc. Preparation of the NDI procedures for insertion in the inspection section shall be in accordance with paragraph 3.6.

3.2 Contents

3.2.1 The manual shall include all nondestructive inspection instructions, procedures, and techniques for the system or support system concerned. NDI techniques shall be developed to inspect critical areas of components or subsystems for potential service defects (cracking, corrosion, wear, or deformation). NDI technique development is required whenever one of the following criteria is met: (a) the NDI method improves safe operation or reliability of the system or subsystem, (b) a saving in maintenance costs or manpower will be realized by using NDI methods, or (c) operational effectiveness or life cycle costs will be favorably effected. Emphasis shall be placed on development of NDI technique that can be applied on the system or subsystem with little or no disassembly (on-condition maintenance). When necessary, multiple NDI procedures may be developed for assembled (on-condition maintenance) and disassembled parts.

3.2.1.1 The manual shall include, but is not limited to procedures covering the following methods: penetrant, magnetic particle, eddy current, ultrasonic, and radiographic inspection. Other methods may be included as necessary if they will provide significant improvement in inspection capability. Proposals for inclusion of methods and equipment not presently used should be submitted during and included in the minutes of the guidance meeting paragraph 4.3.1. Documentation shall be provided showing why existing methods and equipment are not satisfactory and of the improvement to be achieved by the proposed method.

3.2.1.2 Routine visual, magnifying glass, or boroscopic/fiber optic inspections that are normally accomplished by aircraft/power plant mechanics, shall not be included in the NDI manual but shall continue to be included in the applicable maintenance/inspection manual. Special cases may be included at the request of the using command(s) based on component criticallity or of special preparation or inspection technique requirements. Visual or optical inspections may be included in the manual

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3.2.1.3 The manual shall contain detailed step-by-step instructions for each inspection so that a trained NDI technician can duplicate the described inspection. The basic theory and general procedures for use of NDI methods are contained in TO 33B-1-1, and the Nondestructive Testing Handbook. The Nondestructive Testing Handbook shall be used as a guide in the selection and proper application of nondestructive inspection methods. Where the instructions contained in the Handbook differ from those contained in TO 33B-1-1, the latter instructions shall apply.

3.2.2 Verification (Back-up) Inspection Procedures

3.2.2.1 Each inspection will, when necessary, specify a verification (back-up) inspection procedure by another means to verify the initial inspection results. Verification procedures shall be provided in every case where the initial inspection results do not provide uncontestable data for determination of the serviceability of the item inspected.

3.2.2.2 It is desireable to perform the verification procedure by a method employing direct visual observation (optical, magnetic particle, or penetrant) when the initial procedure is performed by an instrumented method (Xray, eddy current or ultrasonic) providing it does not result in extensive disassembly. In those cases where direct observation methods would require considerable disassembly, the use of a different instrumented method is the preferred means of verification. It may be necessary in isolated cases to disassemble for visual confirmation to provide the necessary information for determination of component serviceability.

3.2.3 Illustrations

3.2.3.1 Illustrations to be included in the manual are specified in paragraphs 3.4.2, 3.6, and 3.7. All illustrations shall comply with MIL-M-38784 requirements.

3.2.3.2 Each inspection procedure shall be illustrated with views of the component or subsystem to be inspected showing: (a) general location of the inspection area within the aircraft, missile, engine, or support system; (b) specific location of the component or subsystem to be inspected plus location and orientation of potential defects. This may be shown in two or more views as required for adequate orientation.

3.2.3.3 Symbols that shall be used to represent nondestructive inspection methods and provide detail in illustrating inspection procedures are in Figures 1 & 2. Wherever possible the symbol for the primary and back-up method shall be placed adjacent to the area to be inspected in the next higher assembly illustration

3.2.3.4 Illustrations from existing manuals (Illustrated Parts Breakdown, Maintenance Manuals, etc) shall be used when practical.

3.2.4 Nondestructive Inspection Equipment

3.2.4.1 NDI procedures shall be written in such a manner that Air Force issued equipment (as listed in TA-455) may be used. In those cases where AF standard equipment will not provide the necessary results (ref paragraph 3.2.1.1) other equipment may be used if prior approval is obtained through the procuring activity from the AF NDI Program Manager (SAAMA/MMEW, Kelly AFB, TX).

3.2.4.2 Complete sketches specifying dimensions, surface finish, tolerance, and materials of all special ancillary equipment (standards, shoes, wedges, fixtures, and jigs) shall be included as illustrations. The sketches do not need to be actual size drawings however they shall be adequate to permit future local manufacture by Air Force bases. The sketches shall be provided even when an initial quantity of special ancillary equipment is procured as support equipment.

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3.2.5 Safety. Each inspection procedure included in the manual shall be developed in compliance with MIL-STD-882. Hazards involved, both personal and material, will be clearly annotated and classified per MIL-STD-882. Where required, general safety precautions and specific limitations against the use of NDI equipment on operationally configured aircraft shall be included in Section I, General.

3.2.5.1 Each radiographic technique shall include the following:

WARNING

RADIATION HAZARD

Ensure compliance with all applicable safety precautions set forth in Section 1 of this manual and in TO 33B-1-1. Failure to comply may result in injury to personnel.

3.3 Arrangement. The manual shall be arranged in the following order: Front matter; Introduction; Section 1, General; Section II and subsequent sections. Beginning with Section II, each section shall contain the inspection procedures dealing with a major assembly or subassembly. These sections shall be arranged in the same order as major sections of the Illustrated Parts Breakdown for the aircraft, missile and engine.

3.3.1 Front Matter. Front matter shall consist of: Cover page; list of effective pages; table of contents; list of illustrations; and list of tables. Front matter shall be prepared in accordance with the requirements of MIL-M-38784.

3.3.2 Introduction. The introduction to the manual shall begin with the following text:

1. This manual contains instructions for accomplishing nondestructive inspection of the aircraft (or missile, or engine, etc., as appropriate). The procedures prescribed by this manual are intended to provide step-by-step instructions for the nondestructive inspection of locations where service defects would prevent the items from performing their designated functions. These procedures are developed through maintenance engineering experience and comparison with similar installations on inservice equipment. They are refined and changes as required during the service life of the equipment by continually evaluating the performance of the equipment, results of scheduled inspections, and thorough study of failure data. Local conditions, such as special utilization or climatic environment, may dictate more detailed inspections. Commanders and their maintenance officers are expected to exercise their prerogative to increase the scope of any inspection as required.

2. This manual may pertain to part, or all types and series, of a model, and may, therefore, contain requirements applicable to specific equipment that is not installed on an individual model. When this situation is encountered, those requirements that are not applicable should be disregarded.

3. This manual does not contain inspection level or frequency, acceptance/rejection limitations, or instructions for correcting defective conditions. Inspection levels and frequency are provided in the inspection requirements manuals. Detailed acceptance/rejection criteria and instructions for correcting defective conditions are provided in appropriate maintenance manuals. This information is not contained in this manual since decisions regarding the serviceability of components properly belongs with maintenance technicians trained, skilled, and experienced in their particular speciality, e.g. airframe, hydraulic, propulsion, etc. Also it would duplicate existing information and the task of incorporating the numerous changes to inspection frequency and repair instructions is impractical.

4. The inspection requirements are stated in such a manner as to establish what is to be inspected, what conditions are to be sought, what nondestructive inspection method is to be used, and how the method is to

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be performed. In scope, the inspection procedures are designed to direct attention of maintenance personnel to components and areas where service defects can occur and to provide detailed instructions on the application of nondestructive inspection to assure the integrity of these areas.

5. Nondestructive inspection methods require application by trained, experienced, and proficient technicians. While this manual provides detailed step-by-step procedures for the application of nondestructive methods to inspect specific areas or locations, it must be emphasized that the reliability of the inspection results depends upon the proper application of the readings obtained from the inspection equipment.

6. While using this manual, such adjectives as left and right, upper and lower, front and rear, forward and aft, and clockwise and counterclockwise refer to the airplane (or missile, or engine, etc, as appropriate) as viewed from the rear, on the centerline, looking forward. All dimensions, material gages, and distances are given in inches unless otherwise noted.

7. Changes and supplements to this manual will be published when necessary to add, delete, or change the scope of requirements. Such changes will be based on factual data accumulated as a result of maintenance experience with the equipment. Suggested new or revised field developed inspection procedures shall be prepared on AFTO Forms 242 (Radiographic), 242A (Penetrant), 242B (Magnetic 242C (Eddy Current), 242D (Ultrasonic), Nondestructive Inspection Data per TO 33B-1-1 and submitted as an attachment to AFTO Form 22, Technical Order System Improvement Report per TO 00-5-1.

8. These NDI procedures are directive in nature and deviations without prior approval from the prime AMA limited to compensation of equipment output. Equipment settings, when given, are reference points only, due to the widely varying outputs from different machines. The condition that must be satisfied for accurate inspection is that the inspection equipment be adjusted to obtain the specified response from the set-up or defect standard or the specified density reading on x-ray film. Trained NDI technicians are qualified to make these adjustments.

3.4 Section 1., General

3.4.1 This section shall include, but not be limited to subsections covering: (a) how to use the manual, (b) general description of the aircraft (or missile, or engine, etc), (c) table of related manuals, (d) standard symbols, (e) glossary of special terms, abbreviations, and acronyms, (f) general safety precautions (ref. para 3.2.5), (g) repetitive inspection procedures (e.g. standard penetrant procedures or bolt and bolt-hole inspections), and (h) repetitive inspection equipment set-up or standardizing procedures. Incorporation of repetitive steps in Section 1. will permit reduced text in the detailed inspection procedures by reference to Section 1, however it shall not be cause for deletion of information which is essential for a reliable inspection.

3.4.2 Illustrations and tables in Section 1, General, shall include, but not be limited to: (a) major assembly index, (b) station diagrams, (c) access provisions, (d) standard symbols, (e) illustrations required to clarify repetitive inspections and calibration procedures, and (f) a table containing nomenclature and part, model, or type designator for all required inspection equipment and materials.

3.4.2.1 The illustrations listed in paragraph 3.2.3.2 shall be included in the inspection section of support system manuals (ref para 3.1.2) if they are not contained in other sections of that particular manual.

3.4.3 The first part of Section 1, General, shall consist of short paragraphs defining nondestructive inspection and the five primary nondestructive methods. The intent of these paragraphs is to provide non-technical descriptions for the benefit of individuals who have occasion to utilize the services of nondestructive inspection technicians. General caution notes shall be included relative to: the need for qualified NDI technicians;

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authorization to adjust equipment settings to obtain the necessary response; and radiation hazard warning. Figure 3 is an example of the information desires.

3.5 Subsequent Sections

3.5.1 Each section following Section 1, General, shall contain the inspection procedures for components or items in one major structural group, section, or subsystem. Inspections concerning a particular part or assembly shall be grouped together and each inspection shall be fully developed and documented All instructions must be complete so that any NDI technician can duplicate the inspection with reliable results.

3.5.2 Each section shall contain an index listing the components or parts and the respective inspection procedure. Where practical, this index shall be illustrated with a sketch of the major assembly marked to show the approximate location of the parts to be inspected.

3.5.3 The inspection procedure narration and its accompanying illustrations shall be placed as close together as is possible. The illustration shall be placed behind the narration. When the illustration of a part or component is to be used for two or more procedures, it shall be placed behind the first procedure's narration with the additional procedures following. The procedures shall reference both Figure and page numbers.

3.6 Inspection Procedure

3.6.1 General. The inspection procedure narration shall be arranged in paragraphs and subparagraphs. Inspection procedures shall include pre and post inspection steps (preparation and system securing) that must be accomplished by technicians other than NDI technicians. Wherever possible, these steps shall be summarized by referring to related maintenance manuals (-2-1, -2-2, -23, etc.). These steps shall be expressed in such a manner that they cannot be construed as tasks that must be accomplished by the NDI technician.

3.6.1.1 The required arrangement, format and content of the inspection procedure is provided in the following paragraphs. Examples are provided in figures 4 through 8.

3.6.2 Title. The first paragraph of the inspection procedure shall consist of the nomenclature of the component or subsystem to be inspected. Nomenclature of the component or subsystem to be inspected. Nomenclature shall be consistent with Illustrated Parts Breakdown. Part numbers shall be used as required to positively identify the component. The installation effectivity (when not applicable to all models) of the part or subsystem shall be parenthetically noted after the title, e.g. Stub Longeron, PIN 5B601 (B-52G/H).

3.6.3 Part Description. The second paragraph, with illustration, shall describe and identify the part or component to be inspected. The following information shall be provided in the narration and/or illustration.

3.6.3.1 The part or component to be inspected shall be illustrated per paragraph 3.2.3.2.

3.6.3.2 Narration shall include the material and alloy, heat treat condition, method of manufacture (e.e. casting, forging, extrusion sheet, etc. surface treatments, coatings and finishes.

3.6.3.3 Concise statement or summary of the purpose or Function of the component or part, including the type of loads.

3.6.4 Defect Description. The third paragraph shall be a narrative description of the probable defect and its cause. The defect location shall be cross hatched or otherwise high lighted on the illustrated (ref para 3.6.3 and 3.2.3.2). Orientation of the defect shall be depicted by a distinctive line in the high-lighted area.

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3.6.5 Inspection Procedure. The fourth paragraph shall consist of the title: "PRIMARY METHOD - ULTRASONIC (or other applicable NDI method)". The title shall be capitalized for emphasis. The following subparagraphs, lettered rather than numbered, shall be proceded. Note: The radiation hazard warning, paragraph 3.2.5.1 shall follow the heading when the primary method is radiographic.

3.6.5.1 The first subparagraph shall be titled "NDI Equipment". The paragraph shall list all equipment, materials, standards, shoes, wedges, fixtures, jigs, films, screens, etc., required to perform the inspection. Standard NDI equipment and materials shall be designated by nomenclature and part number, model, or type designator. Illustrations specifying dimensions, and materials of all special standards, shoes, wedges, fixtures, and jigs shall be included (ref para 3.2.3.2). When a standard, shoe, wedge, fixture, or jig is used in more than one inspection, the illustration does not require reprinting. The illustration shall be placed behind the first procedure while subsequent procedures shall refer to the illustration by Figure and Page number.

3.6.5.2 The second subparagraph shall be titled, "Preparation of Aircraft (or Missile, or Engine, as applicable)". The narrative shall describe any preparation necessary, such as: Lower flaps; jack aircraft; remove compressor casing; etc. Wherever possible, the applicable maintenance manual shall be referenced, as:"Remove electronics bay forward rack per TO 1F-4D-2-12 para 6-187." When no preparation is necessary, the statement, "No special preparation required", shall follow the title.

3.6.5.3 The next subparagraph shall be titled "Access". The narrative shall describe how access to the component is gained. This may be as simple as providing the name and location of an access panel or by stating that access is gained with rudder in full right position. In those cases where the item to be inspected is on the exterior and where access is obvious, the subparagraph may be omitted.

3.6.5.4 The next subparagraph shall be titled "Preparation of Part". The narrative shall provide information on what steps are necessary to prepare the part for inspection. This includes any soil removal, degreasing, disassembly, or depainting. Wherever possible, the narrative shall reference the detailed maintenance manuals for explicit instructions. Caution or warning notes, where applicable, relative to possible damage shall be placed ahead of this subparagraph.

3.6.5.5 The next subparagraph shall be titled "NDI Equipment Settings/ Standardization/Set-Up". The narration shall include the specific steps for uniform set-up and standardization of the NDI equipment for the specific inspection. General procedures contained in the equipment operating manuals need not be repeated in these instructions. Repetitive instructions shall be included in Section 1, General and referenced in this subparagraph. Specific data to be included for each NDI method are provided in paragraph 3.7.

3.6.5.7 All inspection procedures shall conclude with the following statement as a subparagraph: "Mark defect(s) and record per TO 00-20-5. See Section 1 for approved marking materials".

3.6.6 Backup NDI Procedure. The fourth numbered paragraph shall describe the verification or backup NDI procedure (ref paragraph 3.2.2). All information required in paragraph 3.6.5 and its subparagraphs shall be provided except the title shall be"BACKUP NDI PROCEDURE". Immediately following the title, the conditions under which the verification procedure is required shall be specified. Subparagraphs that are identical to those in the primary procedure may state, "Same as for Primary Procedure", without repeating the complete subparagraph.

3.6.7 System Securing. The final paragraph in the inspection procedure shall describe the securing and restoration of the component or subsystem and aircraft (or missile, or engine, etc). This paragraph is intended to provide only general requirements and shall reference the applicable maintenance manual for detailed instructions pertaining to restoration

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of the system to an operating configuration.

3.7 Specific NDI Method Data Requirements.

3.7.1 General. The following minimum information shall be provided in each inspection procedure as outlined in paragraph 3.6, Inspection Procedure. All information shall be presented in step-by-step narrative form with illustrations for clarity. Standard NDI symbols (attachment 1) shall be used to show NDI equipment placement and usage.

3.7.2 Penetrant Method Requirements.

Inspection procedure
1. Pre inspection depainting and cleaning
2. Penetrant material type, sensitivity, and limitation
3. Method of application
4. Dwell times for penetrant, emulsifiers, and developers, including temperature at which dwell times were developed
5. Lighting requirements
6. Magnification (if required)
7. Post inspection cleaning

3.7.3 Magnetic Particle Method Requirements

Inspection procedure
1. Removal of all magnetic sensitive equipment from area being inspected.
2. Pre inspection cleaning
3. Method and type of magnetic field (longitudinal/circular/central conductor/AC or DC).
4. Method, type, concentration, and color of inspection media (wet/dry/continuous/residual and powder color).
5. Method of media application (if pump and hose not used).
6. Direction of magnetic flux flow. Illustration shall show the magnetic field in relation to the defect.
7. Amperes of magnetizing current (meter reading) or ampere turns of current required. Time and number of shots. Number and size of coils or cable loops used for longitudinal magnetization.
8. Directions for placing item in contact, loop or coil size, placement and spacing of probe legs, placement or location of the coil.
9. Location and cause of false indications (if required).
10. Magnification (if required)
11. Demagnetization procedures, including procedure for check level of residual magnetization. Include acceptable level of residual magnetication.
12. Post inspection cleaning.

3.7.4 Eddy Current Method Requirements.

NDI Equipment Settings/Standardizing/Setup
1. Refer to Section 1, General, For equipment setup procedure including adjustment for sensitivity and liftoff.
2. Step-by-step procedure to obtain desired meter deflection from defect standard or set up block. Instructions for manufacture of standard or block shall be included per para 3.2.4.2.

Inspection procedure
1. Area to be scanned, number of passes, probe translation, and scan rate (narrative and illustration(s) showing inspection are and probe placement).
2. Defect response and possible nonrelevant indications, such as edge effect, fasteners, etc.

3.7.5 Ultrasonic Method Requirements.

NDI Equipment Settings/Standardizing/Set up
1. Refer to Section 1, General For initial equipment setup and sensitivity levels to provide a starting point for subsequent

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adjustment.
2. Step-by-step procedure for equipment adjustment to produce desired response from defect standard. Illustrated, as required, to show adjustment of sweep delay, sweep length, markers, and response amplitude without reference to dial settings.
3. Illustrations of the cathode ray tube presentation must show response signals in relation to transducer location. Changes in response signals with transducer scanning must be illustrated or described.
4. Instructions for manufacture of standard shall be included per para 3.2.4.2. The standard shall reflect a sensitivity level adequate to detect size considered rejectable by the applicable technical manual.

Inspection procedure
1. Area to be scanned, application and type of couplant, transducer movement, defect response and possible nonrelevant indications. (Illustration (s) shall show defect response and nonrelevant indications correlated to transducer positions).
2. Post inspection cleaning.

3.7.6 Radiographic Method Requirements.

NDI Equipment Settings/Standardizing/Setup.
1. Description of film, class or type based on signal-to-noise ratio (TO 33B-1-1), size, and type of packaging, i.e. day-pack, lead pack, or cassettes.
2. When more than one film is to be exposed, the number shall be stated and film placement shown in the illustration. When multiple film technique is used, enter the film type nearest the Xray tube, a slash mark,followed by the second type film.
3. When special cut film or odd shapes are to be used, a dimensioned sketch of the film shall be provided.
4. The following information shall be provided on the illustration (s): film placement on the component or subsystem to be inspected; aiming point plus two dimensions for location; and location of the source. Three dimensions are required for source location including the focal point to film or focus point to aiming point. Angles shall not be used. All reference points for locating the film and tube head must be easily recognized on typical service configured aircraft, missile, or engine. Stenciled reference points (body stations, exterior bulk-head lines, etc.) and rivet or other flush fasteners may not be recognizable on in-service systems.
5. Reference equipment settings, i.e. Kilovoltage, milliamperage, and exposure time. Indicate whether film was hand or automatic processed and type of processing chemicals used at these equipment settings.
6. Required density in H & D units and area on film where density is measured.
7. If required, penetrameter location.

Inspection procedure
1. Step-by-step inspection procedure.
2. If required, special film reading requirements (light intensity, magnification,etc).

4. Quality Assurance Provisions.

4.1 Production Quality Control. The contractor shall maintain a quality control program which is adequate to insure the end product meets the requirements of this specification and MIL-M-38784. The Government reserves the right to review and perform any inspections deemed necessary to assure the manual conforms to prescribed requirements.

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4.2 Verification of Procedures. The contractor may develop NDI procedures in the test laboratory using mock-up or simulated components or parts. However, all inspection procedures shall be verified on a service configured aircraft (or missile, or engine, etc) prior to the final draft of the procedure. Following verification, the proposed procedure shall be reviewed by the contractor's engineers.

4.3 Guidance, In-process and Prepublication Review Meetings.

4.3.1 Guidance Meeting. The contractor shall schedule a meeting following contract award and prior to starting work on the NDI manual. The purpose of this meeting is to promote a better understanding between the Air Force and the contractor of their respective obligations by exchange of viewpoints through personal discussion. This interchange of ideas will familiarize the contractor with Air Force field and depot maintenance procedures and will provide Air Force participants with a detailed knowledge of the contractor's proposed work effort. Information obtained through this verbal interchange is not possible to obtain through written communication. This meeting and the subject matter shall not modify or change in any manner any of the existing contractual obligations except as authorized by the Contracting Officer in writing. The contractor shall notify the procuring activity and SAAMA/MMEW-4 at least twenty one days prior to the actual meeting of the scheduled date and location.

4.3.2 In-Process and Prepublication Review Meetings. In-process and prepublication review meetings shall be held in accordance with MIL-M-38784. The contractor's proposed schedule for in-process and prepublication review meetings shall be an agenda item at the guidance meeting.

5. Preparation For Delivery.

5.1 Packaging, packing, and marking for shipment shall be in accordance with MIL-M-38784 and MIL-P-38790.

6. Notes.

6.1 Intended Use. This specification is to be used in the preparation of NDI manuals covering aircraft, missiles, engines, and their supporting systems (see paragraph 1).

6.2 Definitions.

6.2.1 Nondestructive Inspection (NDI). A family of methods for investigating the quality , integrity, properties and dimensions of materials and components without damaging or impairing their serviceability.

6.2.2 On Condition Maintenance. Application of inspection and testing procedures and tichniques without removal or disassembly that allows the condition of the equipment to dictate the need for maintenance or the extent of the repair/overhaul required to restore serviceability.

6.2.3 NDI Technician. Individuals who have been trained in NDI (completed the 3 or 5 level ATC NDI training course or equivalent per TO 33B-1-1) and have demonstrated a satisfactory degree of NDI proficiency.

6.2.4 Critical Area. Determination of critical areas or components is based on (a) safety of flight considerations (primary structure and non-redundant subsystem components the failure of which could cause loss of the aircraft, missile, or crew), (b) mission essential considerations (secondary structure and subsystem components the failure of which would degrade the ability of the aircraft or missile to perform its mission), or (c) maintenance economy considerations (structure and subsystem components that would require extensive or repetitive disassembly to perform visual inspection of internal areas). Critical areas are not necessarily limited to structural components; all subsystem components must be analyzed for criticallity.

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6.2.5 Discontinuity. Break or interruption in the normal physical structure of a material or component. Examples are cracks, voids, inclusions, and laminations.

6.2.6 Defect. A discontinuity or dimensional change which degrades the serviceability of the part.

6.2.7 Nonrelevant Indication. An equipment response that is due to factors other than discontinuities or defects.

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Air Force - 16

Preparing Activity:

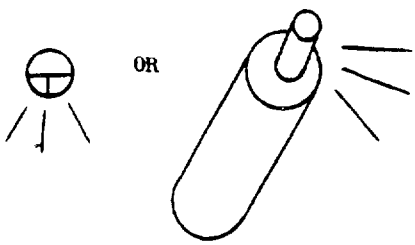
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Review Activities:
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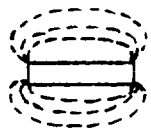
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NDI METHOD IDENTIFICATION SYMBOLS



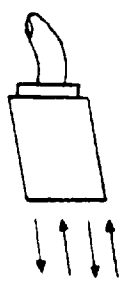
RADIOGRAPHIC



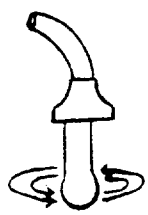
MAGNETIC PARTICLE



PENETRANT



ULTRASONIC



EDDY CURRENT

Figure 1. Symbols Used to Indecate the Primary and Verification Method of Accomplishing Nondestructive Inspection.

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NDI PROCEDURE ILLUSTRATION SYMBOLS

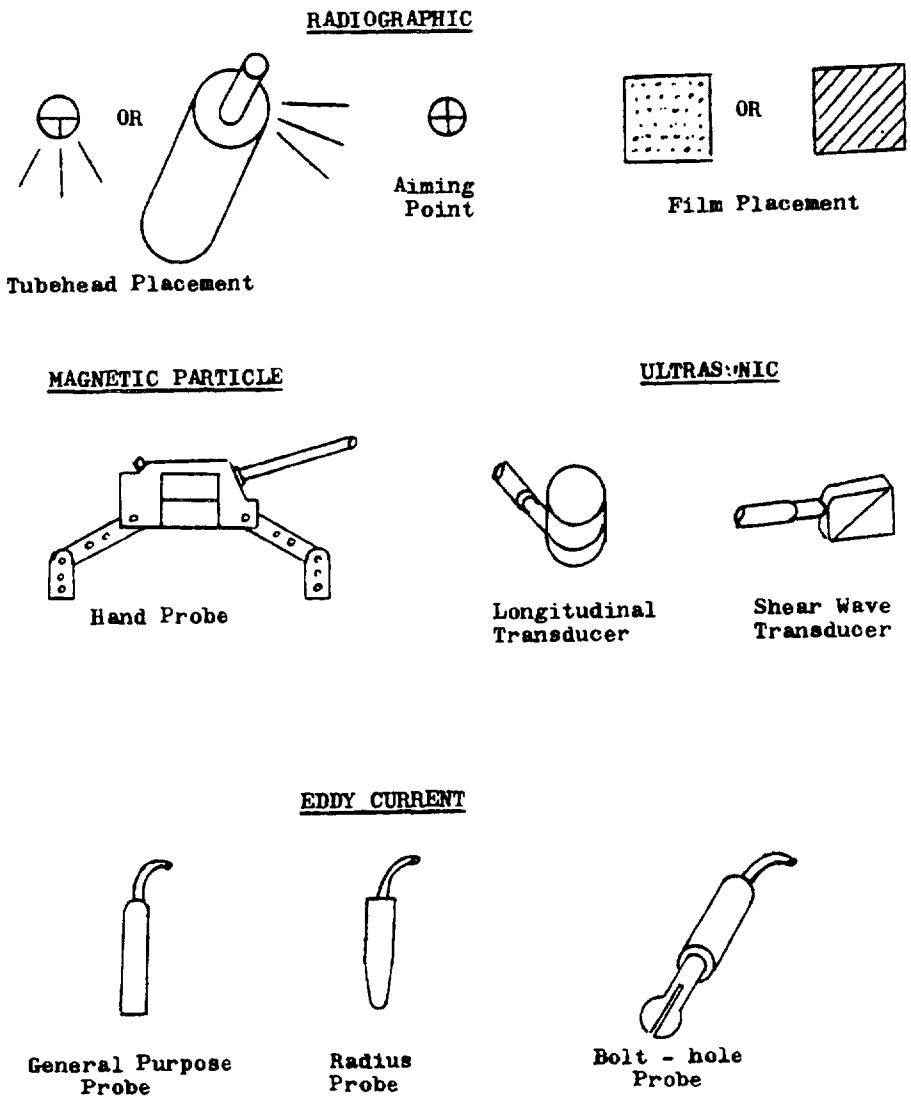


Figure 2. Symbols Used to Provide Detail in Illustrating Nondestructive Inspection Procedures

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SECTION I
GENERAL

1-1. NONDESTRUCTIVE INSPECTION.

1-2. A family of methods for investigating the quality, integrity, properties, and dimensions of materials, and components, without damaging or impairing their serviceablilty, by use of penetrant, magnetic particle, eddy current, ultrasonic, radiographic, etc, inspections.

NOTE

Effective utilization of NDI can be attained only by trained and experienced personnel.

1-3. NONDESTRUCTIVE INSPECTION METHODS.

1-4. PENETRANT INSPECTION METHOD. (Refer to TO 33B-1-1, Chapter 6.). Penetrant inspection is a nondestructive test for discontinuities open to the surface in parts made of nonporous materials. This is done by treating the whole area with a fluid which penetrates into the surface discontinuity. The surplus penetrant, not in the discontinuity is removed and the penetrant remaining in the discontinuity is returned to the surface by capillary action. A developer is applied to provide a contrasting surface and through absorption forms an indication large enough to be noticeable to the eye. It is a quick and positive method for detecting many different types of surface discontinuities. Its chief limitations are that discontinuities must be open to the surface and it is essential that there be no contaminating material around or in the discontinuity which might either prevent the penetrant from entering the defect or which may reduce its visibility. Visible dye penetrants have the lowest sensitivity of any penetrant and are not to be used for the inspection of service defects on aircraft, missile or engine components.

1-5. MAGNETIC PARTICLE INSPECTION METHOD (Refer to TO 33B-1-1, Chapter 2). Magnetic particle inspection is a rapid nondestructive means of detecting discontinuities in materials having ferromagnetic properties, principally iron and steel. Magnetic particle inspection is accomplished by inducing a magnetic field into a part. A defect will interrupt this field creating new north and south poles with the magnetic flux bridging the defect. This flux liakage attracts the magnetic particles to form the indication. In addition to cracks, the types of flaws that can be detected include seams, laps, folds, and nonmetallic inclusions that are either surface or slightly subsurface.

1-6. EDDY CURRENT INSPECTION METHOD (Refer to TO 33B-1-1, Chapter 2). Eddy current inspection is a nondestructive means of obtaining information a number of material variables such as alloy type, hardness, heat treat condition, thickness, cracks, etc. An insulated coil induces eddy currents by being placed in close proximity to a conductive test piece. Surface or near surface discontinuities cause a major disruption of the eddy current field, thus producing a detectable change. These changes can be read out on a meter, cathode ray tube, or recorder.

Figure 3. Example of Information Desired in Section I. (Sheet 1 of 2)

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1-7. ULTRASONIC INSPECTION METHOD (Refer to TO 33B-1-1, Chapter 4). Ultrasonic inspection is a method of inspection, using sound waves above the audible range. High frequency sound is induced into the part by a transducer. This ultrasonic energy travels through the part. Any marked changes in acoustic properties (flaw,interface or back surface) will reflect the sound back to the transducer. This information is normally displayed on an oscilloscope. Proper selection of transducer, frequency, sensitivity, angle, etc, will enable inspection of the surface, subsurface, and back surface of the part.

1-8. RADIOGRAPHIC INSPECTION METHOD (Refer to TO 33B-1-1, Chapter 5). Radiographic inspection is a nondestructive inspection method utilizing a source of X-rays or gamma rays to detect discontinuities in materials and components. X-rays penetrate metallics and nonmetallics and are differentially absorbed. Discontinuities, which are less dense than metal absorb less radiation and thus are shown on the recording medium as dark shadows. The recording medium, for the purposes of this manual, is film but it can be any means that converts radiation energy into a visible image. Radiography is used to inspect both metallic and non metallic materials. It has the capability of inspecting the interior of opaque objects or assemblies without access to the inside. However, the defects must be aligned and oriented with respect to the X-ray beam and film. Radiography is one of the most expensive and seast sensitive test methods for fine crack detection and should only be used to detect flaws which are not accessible or improperly oriented for the use of other test methods.

NOTE

There is considerable variation in equipment output with respect to numerical settings. NDI technicians are required to compensate for the variations to produce the necessary response or densities. (Refer to TO 33B-1-1).

1-2

Figure 3. Example of Information Desired in Section I. (Sheet 2 of 2)

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6-156. VISOR DOOR BULKHEAD RING, INNER LEG, FUSELAGE STATION 281, DRAWING 4F73790.

6-157. DESCRIPTION. (See figure 6-28-6/0.) The visor door bulkhead ring is located at fuselage station 281. The bulkhead ring is made from a 7075-T6 aluminum extrusion and is given a chemical conversion coating and two coats of zinc chromate primer. The ring is attached to the bulkhead panel, has beam and intercostal attachments, traverses several panel splices and has seven splices of its own.

6-158. DEFECTS. Overload stresses may concentrate at ring and panel splices and at the beam attach regions, possibly resulting in cracks appearing in the frame fillet radius, as illustrated in figure 6-29-6/0.

6-159. PRIMARY NDI PROCEDURE - RADIOGRAPHIC.

WARNING

RADIATION HAZARD

Ensure compliance with all applicable safety precautions set forth in Section I of this manual and in T.O. 33B-1-1. Failure to comply may result in injury to personnel.

a. NDI equipment:

(1) X-ray Machine, Portable, 160KVP, 40 degree tube head, Sperry P/N SPX-160E-8-C or equivalent.

b. Preparation of airplane: Remove the nose radome and insulation blankets, in accordance with T.O. 1C-5A-2-2, Section II. Remove the forward ramp extension in accordance with T.O. 1C-5A-2-12, Section II.

WARNING

Prior to approaching the visor door, disable all visor door operating systems in accordance with T.O. 1C-5A-2-12, Section II. Failure to comply may result in injury to personnel.

c. Access: Inspection areas are accessible from inside the cargo compartment with forward ramp in normal closed position. Inspection will require the use of AGE platforms for elevation inside and outside the airplane.

d. Preparation of part: No special preparation required.

e. X-ray machine settings: Set X-ray machine for technique indicated in Radiographic Inspection Data Block in figure 6-28-6/0. This procedure is for a film density of 1.5 to 2.5.

Note

Refer to Section I for film markings/location instructions.

f. Film placement: Place X-ray film on the forward side of the visor door bulkhead ring inner leg as illustrated in figure 6-28-6/0 for each exposure. After an exposure, remove the exposed film and place film for the next exposure. Use film specified in Radiographic Inspection Data in figure 6-28-6/0.

g. X-ray tube location: Locate the X-ray tube as illustrated in figure 6-28-6/0 for each exposure. After exposing each film, relocate the tube for the next exposure, as illustrated in figure 6-28-6/0.

h. Inspection: Make radiographs of the visor door bulkhead ring inner leg in accordance with the numbered exposures illustrated as indicated in the Radiographic Inspection Data in figure 6-28-6/0.

i. Mark and report indicated defects. See Section I for approved temporary marking materials.

6-160. BACKUP NDI PROCEDURE - None Required.

6-161. SYSTEM SECURING. Clean areas inspected, restore finishes, reinstall removed components, and perform operational checkouts, as required, in accordance with applicable T.O.'s.

Figure 4. Example of Format and Content of Radiographic Inspection Procedures. (Sheet 1 of 5)

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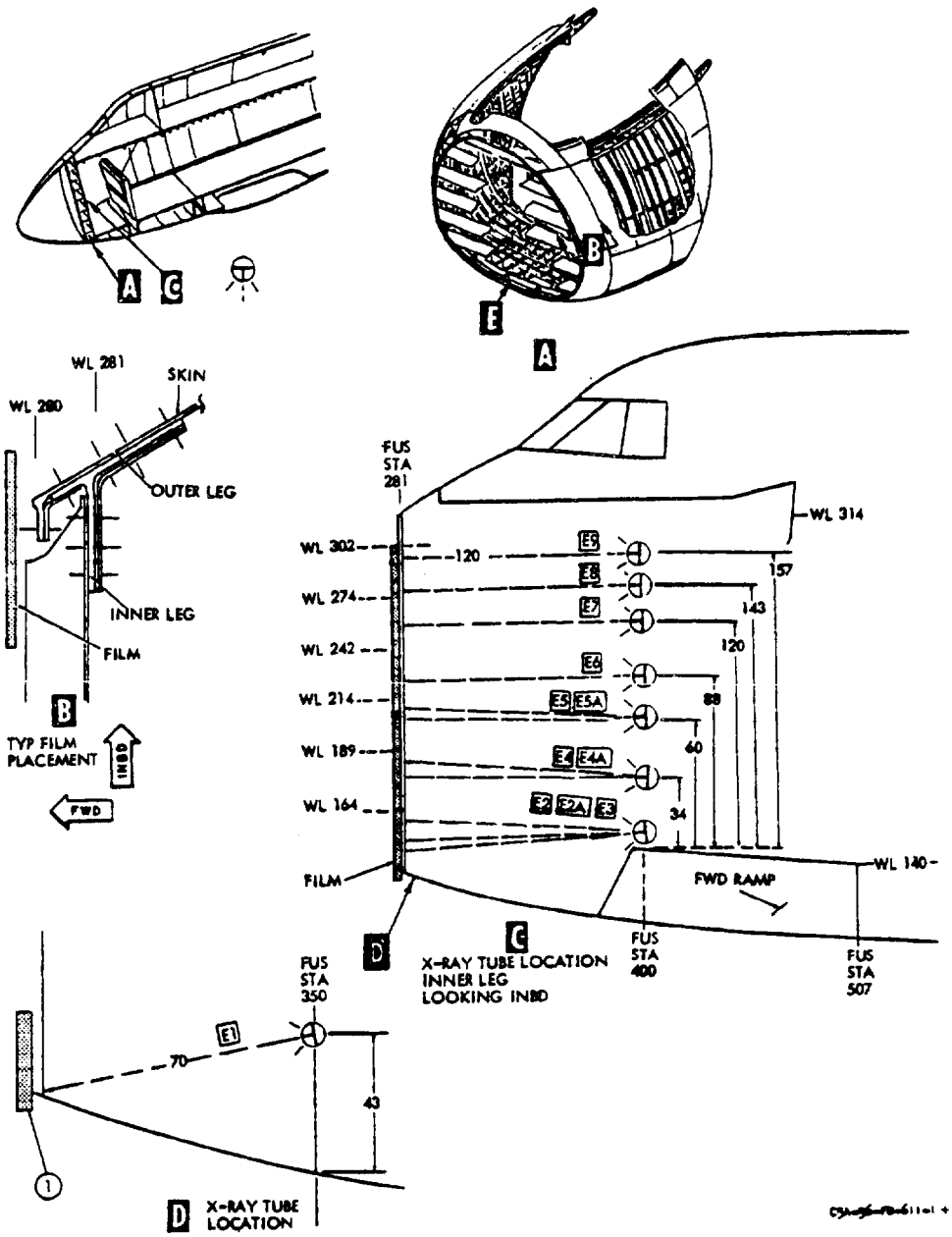


Figure 6-28-6/0. Visor Door Bulkhead Ring, Inner Leg, Fuselage Station 281 (Sheet 1 of 4)

6-73

Figure 4. Example of Format and Content of Radiographic Inspection Procedures. (Sheet 2 of 5)

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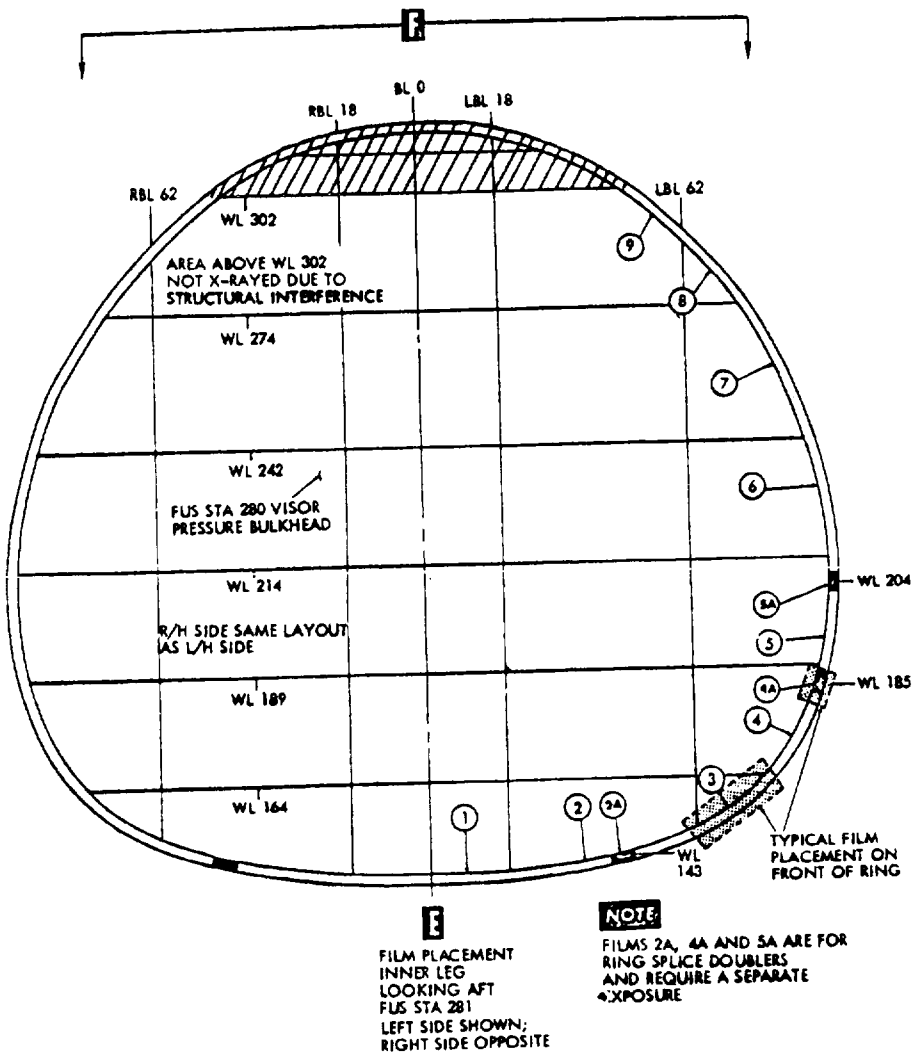
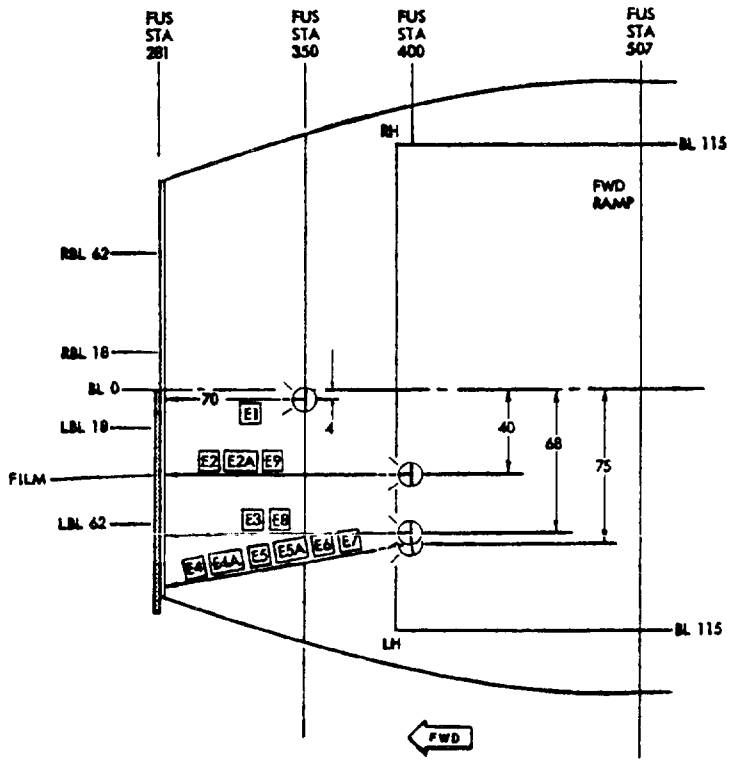


Figure 6-28-6/Q. Visor Door Bulkhead Ring, Inner Leg, Fuselage Station 281 (Sheet 2 of 4)

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Figure 4. Example of Format and Content of Radiographic Inspection Procedures. (Sheet 3 of 5)

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Figure 6-28-4/0. Visor Door Bulkhead Ring, Inner Leg, Fuselage Station 281 (Sheet 3 of 4)

6-75

Figure 4. Example of Format and Content of Radiographic Inspection Procedures. (Sheet 4 of 5)

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RADIOGRAPHIC INSPECTION DATA							
EXPOSURE NUMBER	KV	MA	FFD (INCHES)	TIME (SEC)	FILM		REMARKS
					TYPE	SIZE	
E1	130	4	70	180	B	7 X 24	FILM NO. SAME AS EXPOSURE NO.
E2	150	4	120	180	B	7 X 50	
E2A	150	4	120	210	A	10 X 12	
E3	150	4	120	180	B	7 X 30	
E4	150	4	120	180	B	7 X 34	FILM SIZE IS FOR ROLL FILM. 7X17 REGULAR SIZE FILM MAY BE USED IF DESIRED.
E4A	150	4	120	210	A	10 X 12	
E5	150	4	120	180	B	7 X 32	
E5A	150	4	120	210	A	10 X 12	
E6	150	4	120	180	B	7 X 34	
E7	150	4	120	180	B	7 X 40	
E8	150	4	120	180	B	7 X 28	
E9	150	4	120	180	B	7 X 24	

C5A-36-FB-6-1 sh 4

Figure 6-28-6/0. Visor Door Bulkhead Ring, Inner Leg, Fuselage Station 281 (Sheet 4 of 4)

6-76

Figure 4. Example of Format and Content of Radiographic Inspection Procedures. (Sheet 5 of 5)

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6-222. VISOR DOOR SIDE LATCH HOOKS, WATER LINE 306, DRAWINGS 4F73300, 4F73710, 4F73712, 4F73714, 4F73716, 4F73718, 4F73720, 4F73722 AND 4F73724.

6-223. DESCRIPTION. (See figure 6-40-2/0.) There are sixteen latch hooks located in the forward fuselage, eight hooks each located on the left and right sides at Fuselage Station 310, 330, 350, 375, 398, 416, 443 and 464, and Water Line 306. The latch hook is made of PH13-8Mo steel, heat treated to 200 KSI. The hook has a serrated back face and slotted holes which allow for mating with the serrated plate and satisfactory positioning of this assembly when hook and spool are engaged.

6-224. DEFECTS. Possible misalignment of the visor could cause excessive loading on the hook, leading to cracking inside the hook radius.

6-225. PRIMARY NDI PROCEDURE - FLUORESCENT PENETRANT.

a. NDI equipment:

(1) Fluorescent penetrant materials and equipment selected from approved list in table 1-III.

b. Preparation of airplane: Open the visor door in accordance with the instructions contained in T.O. 1C-5A-2-12, Section II.

c. Access: Inspection areas are accessible with the visor door open. Use AGE platforms for elevation.

CAUTION

High strength steels are extremely susceptible to hydrogen embrittlement caused by acid-base cleaners and solvents. Use only cleaners and solvents that are certified safe for use on high strength steel parts. Failure to comply may cause cracks in parts.

d. Preparation of part: See T.O. 1C-5A-23, Section V, for finish removal/restoration and cleaning instructions. Strip finish from logcal suspected defective areas. Clean local inspection areas.

e. Instrument settings/calibration: Not applicable.

f. Inspection: Perform fluorescent penetrant inspection in accordance with table 1-I, Section I, and instructions contained in T.O. 33B-1-1.

g. Mark and report indicated defects. See Section I for approved temporary marking materials.

6-226. BACKUP NDI PROCEDURE. None required.

6-227. SYSTEM SECURING. Clean areas inspected, restore finishes, reinstall removed components, and perform operational checkouts, as required, in accordance with applicable T.O.'s.

Figure 5. Example of Format and Content of Fluorescent Penetrant Inspection Procedures. (Sheet 1 of 2)

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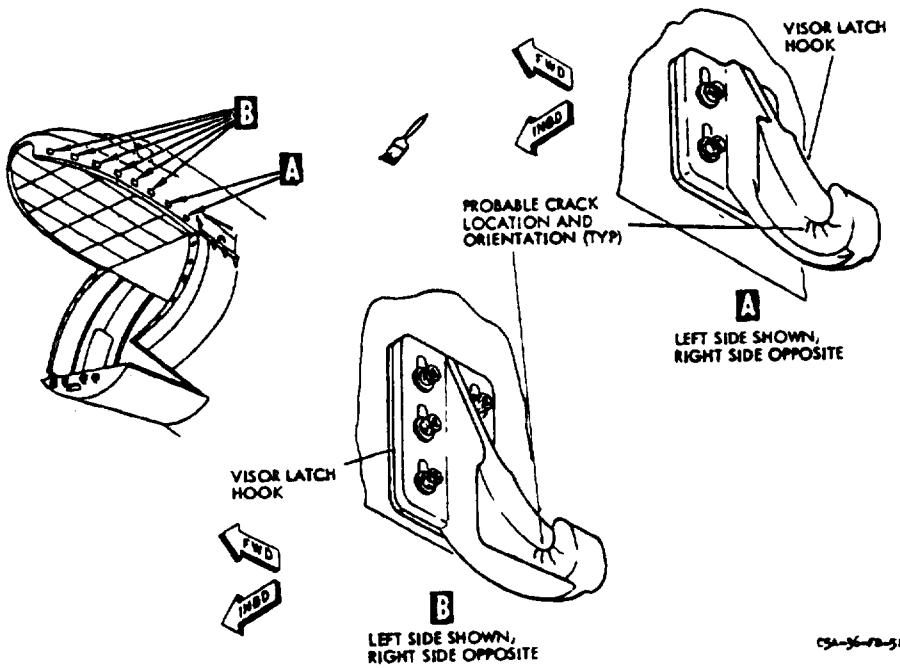


Figure 6-40-2/0. Visor Door Side Latch Hooks, Water Line 306 6-121

Figure 5. Example of Format and Content of Fluorescent Penetrant Inspection Procedures. (Sheet 2 of 2)

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6-373. FORWARD RAMP EXTENSION AFT BEAM HINGE, DRAWINGS 4F52111 AND 4F52817.

6-374. DESCRIPTION. (See figure 6-70-3/0.) Eight machined high strength steel hinges attach the forward ramp extension to the forward ramp structure. These hinges, located on the aft beam of the forward ramp extension, are cadmium-titanium alloy plated and painted with zinc chromate primer. The 4F52111 hinges are made from 4340 alloy steel heat treated to 200 - 220 KSI ultimate tensile strength, and are used on airplanes serialized through AF 67-174. The 4F52817 hinges, found on airplanes serialized AF 68-211 and up, are manufactured from 17-4 PH stainless steel heat treated to 190 KSI ultimate tensile strength. The two types of hinges are similar in configuration.

6-375. DEFECTS. Eccentric loading of the ramp and ramp extension during service operations could cause cracks to propagate from hinge attach fastener holes.

Note

Two primary NDI Procedure are provided; No. 1 for the 4340 alloy steel hinges, and No. 2 for the 17-4PH stainless steel hinges.

6-376. PRIMARY NDI PROCEDURE NO. 1 - MAGNETIC PARTICLE.

a. NDI equipment:

(1) Magnetic Inspection Unit, 144-inch, Type MB-3, 6000 amperes, or equivalent.

(2) Fluorescent Magnetic Particle Materials.

b. Preparation of airplane: Remove forward ramp extension from the airplane in accordance with T.O. 1C-5A-2-12, Section II. Remove eight hinges from the ramp extension aft of beam in accordance with T.O. 1C-5A-2-12, Section I.

c. Access: Inspection areas are accessible with removals described in step b above accomplished.

CAUTION

High strength steels are extremely susceptible to hydrogen embrittlement caused by acid-base cleaners and solvents. Use only cleaners and solvents that are certified safe for use on high strength steel parts. Failure to comply may cause cracks in parts.

d. Preparation of part: See T.O. 1C-5A-23, Section V, for cleaning instructions. Clean hinges of contaminants that could interfere with concentration of magnetic particles at discontinuities or cause improper accumulation of particles.

e. Instrument settings/calibration: Calibration not applicable. Set selector switch on unit for direct current power supply. Adjust current as follows:

(1) Circular field: 1600 - 2000 amperes.

(2) Longitudinal field: 1600 - 2000 amperes, 5-turn coil.

f. Inspection: Inspect hinges by the wet continuous magnetic particle method, utilizing both circular and longitudinal magnetic fields, in accordance with T.O. 33B-1-1. Demagnetize parts upon completion of inspection.

g. Mark and report indicated defects. See Section I for approved temporary marking materials.

6-377. PRIMARY NDI PROCEDURE NO. 2 - FLUORESCENT PENETRANT FOR 17-4PH STAINLESS STEEL.

a. NDI equipment:

(1) Fluorescent penetrant materials and equipment selected from approved list in table 1-III.

b. Preparation of airplane: Same as for Primary NDI Procedure No. 1.

c. Access: Same as for Primary NDI Procedure No. 1.

CAUTION

High strength steels are extremely susceptible to hydrogen embrittlement caused by acid-base cleaners and solvents. Use only cleaners and solvents that are certified safe for use on high strength steel parts. Failure to comply may cause cracks in parts.

d. Preparation of part: See T.O. 1C-5A-23, Section V, for finish removal/restoration instructions. Strip finish from local suspected defective areas. Clean local inspection areas.

e. Instrument settings/calibration: Not applicable.

f. Inspection: Perform fluorescent penetrant inspection procedure per table 1-II in Section I in accordance with instructions contained in T.O. 33B-1-1.

g. Mark and report indicated defects. See Section I for approved temporary marking materials.

6-378. BACKUP NDI PROCEDURE: None required.

6-379. SYSTEM SECURING. Clean areas inspected, restore finishes, reinstall removed components, and perform operational checkouts, as required, in accordance with applicable T.O.'s.

6-230

Figure 6. Example of Format and Content of Magnetic Particle Inspection Procedures. (Sheet 1 of 2)

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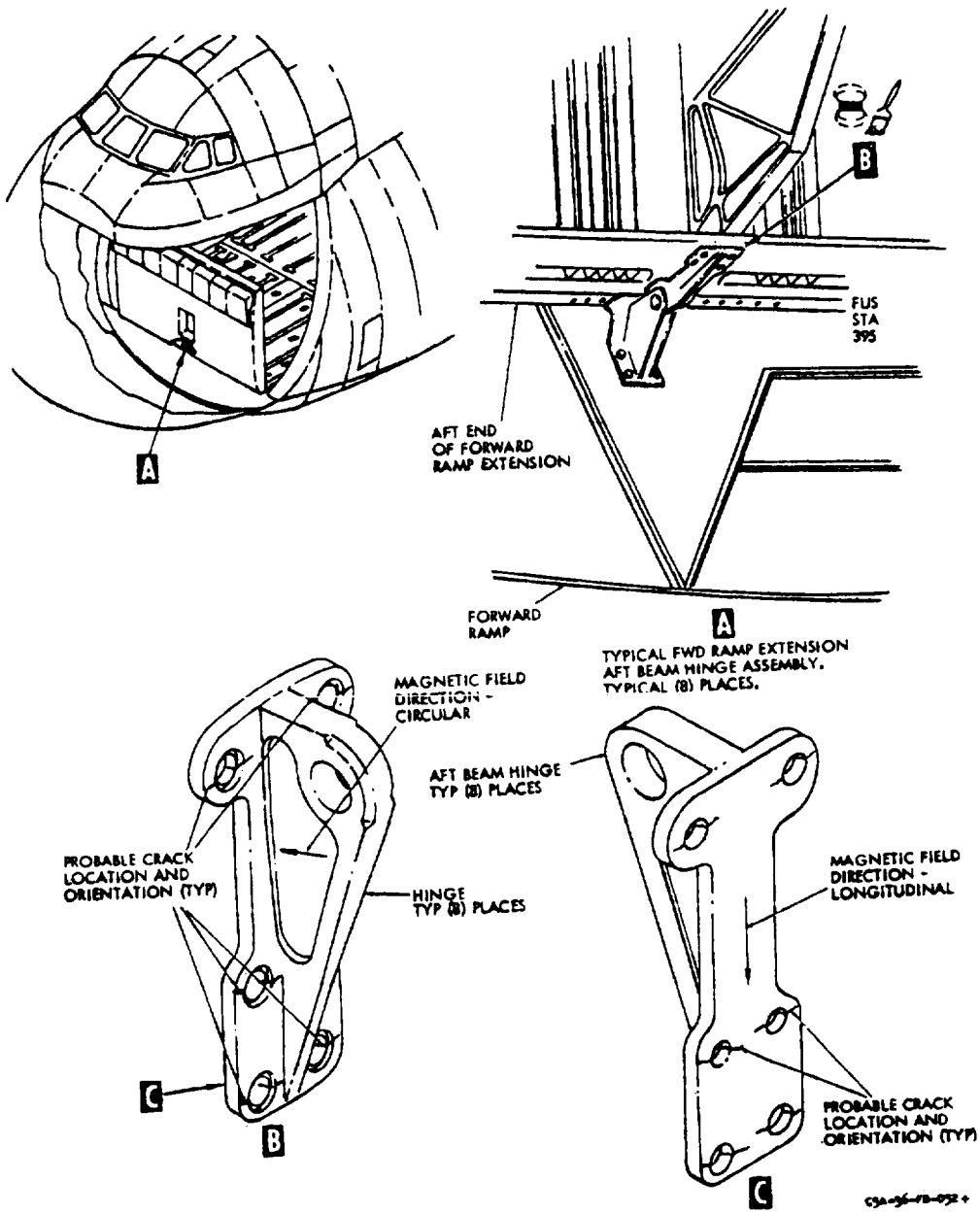


Figure 6-70-3/0. Forward Ramp Extension Aft Beam Hinge 6-331

Figure 6. Example of Format and Content of Magnetic Particle Inspection Procedures. (Sheet 2 of 2)

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6-380. FORWARD RAMP EXTENSION LONGITUDINAL BEAM AND AFT HINGE FITTING ATTACH POINTS AND ACTUATOR ATTACH POINTS, DRAWINGS 4F52135, 4F52136, AND 4F52137.

6-381. DESCRIPTION. (See figure 6-71-4/2.) The eight forward ramp extension aft hinge fittings are attached to the upper rear areas of the ramp extension longitudinal beams. Two of the hinge fittings serve as anchor points for the forward ramp extension actuator fittings. The beams are machined from 7075-T651 aluminum alloy, are sulfuric acid anodized and painted with two coats of zinc chromate primer.

6-382. DEFECTS. The attach points on the beams are exposed to atmospheric conditions, and high stresses during airplane loading and unloading operations. Cracks may originate radiating from the truss bar attach bolt hole in two of the beams and from three nut retaining holes in each beam.

6-383. PRIMARY NDI PROCEDURE - EDDY CURRENT.

a. NDI equipment:

(1) Crack Detector, Magnaflux ED-520 or equivalent.

(2) Probe, Bolt Hole, 3/8-inch to 1/2-inch diameter, Magnaflux P/N 200852 or equivalent.

(3) Probe, Bolt Hole, 1/2-inch to 5/8-inch diameter, Magnaflux P/N 200853 or equivalent.

(4) Probe, Bolt Hole, 5/8-inch to 1-inch diameter, Magnaflux P/N 200854 or equivalent.

b. Preparation of airplane: Open visor door in accordance with T.O. 1C-5A-2-12, Section II.

c. Access: Inspection areas are accessible with removals described in step b, and visor door open.

d. Preparation of part: Remove forward ramp hinges, one at a time, and attachment barrel nuts and actuator strut (2 places) in accordance with T.O. 1C-5A-2-12, Section II. Remove strut and actuator attach fitting per T.O. 1C-5A-3, Section IV. See T.O. 1C-5A-23, Section V, for cleaning instructions. Clean local inspection areas as necessary to permit good contact between part and probe.

e. Instrument settings/calibration: See paragraph titled Eddy Current Instrument Calibration in Section I.

Note

When changing inspection probes, recalibrate eddy current instrument.

f. Inspection: Using appropriate bolthole probe, scan inside the holes from the accessible sides of longitudinal beams as illustrated. A sharp meter deflection, as noted during calibration using calibration standard in accordance with Section I, will indicate a probable crack radiating from the hole.

g. Mark and report indicated defects. See Section I for approved temporary marking materials.

h. To confirm defect conditions, proceed with Backup NDI Procedure. If no defect is suspected, proceed to System Securing paragraph.

6-384. BACKUP NDI PROCEDURE - FLUORESCENT PENETRANT. Perform fluorescent penetrant inspection of suspected defective areas as required.

a. NDI equipment:

(1) Fluorescent penetrant materials and equipment selected from approved list in table 1-III.

b. Preparation of airplane: Same as for Primary NDI Procedure.

c. Access: Same as for Primary NDI Procedure.

d. Preparation of part: Remove forward ramp hinges, one at a time, and attachment barrel nuts and actuator sturt (2 places) in accordance with T.O. 1C-5A-2-12, Section II. Remove strut and actuator attachment fitting per T.O. 1C-5A-3, Section IV. See T.O. 1C-5A-23, Section V, for finish removal/restoration instructions. Strip finish from local suspected defective areas. Clean local inspection areas as required in accordance with T.O. 1C-5A-23, Section V.

e. Instrument settings/calibration: Not applicable.

Note

If a suspected crack is believed to originate at a fastener hole, remove the fastener (eddy current bolt hole inspection may be used) and strip and clean surface under fastener prior to penetrant inspection.

f. Inspection: Inspect only the suspected defective areas. Perform fluorescent penetrant inspection in accordance with table 1-I, Section I, and instructions contained in T.O. 33B-1-1.

g. Mark and report indicated defects. See Section I for approved temporary marking materials.

6-385. SYSTEM SECURING. Clean areas inspected, restore finishes, reinstall removed components and perform operational checkouts, as required, in accordance with applicable T.O.'s.

6-232

Figure 7. Example of Format and Content of Eddy Current Inspection Procedures. (Sheet 1 of 3)

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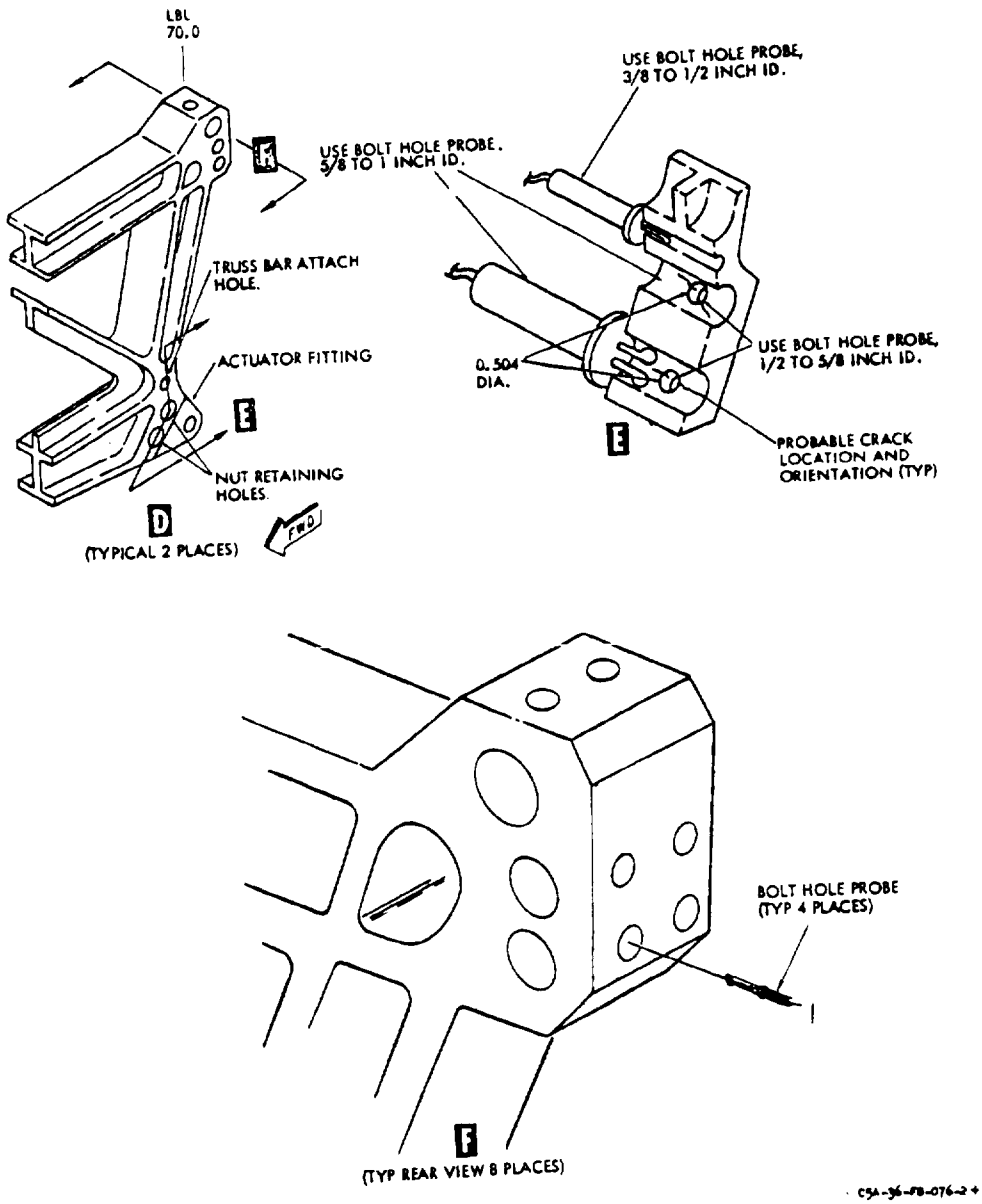


Figure 6-71-4/2. Forward Ramp Extension Longitudinal Beam and Aft Hinge Fitting Attach Points and Actuator Attach Points (Sheet 2 of 2)

6-234

Figure 7. Example of Format and Content of Eddy Current Inspection Procedures. (Sheet 3 of 3)

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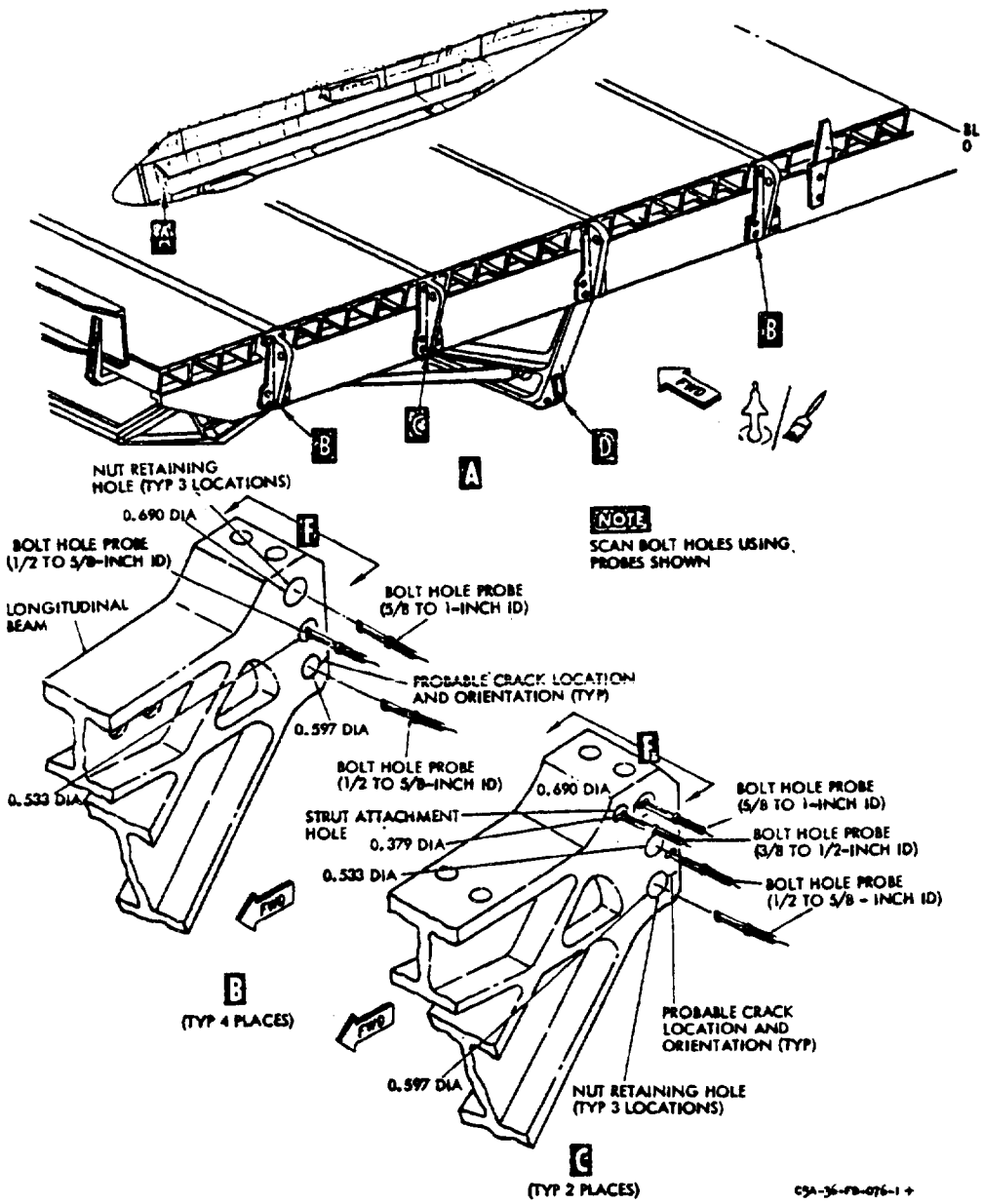


Figure 6-71-4/2. Forward Ramp Extension Longitudinal Beam and Aft Hinge Fitting Attach Points and Actuator Attach Points (Sheet 1 of 2)

6-233

Figure 7. Example of Format and Content of Eddy Current Inspection Procedures. (Sheet 2 of 3)

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6-852. MAIN LANDING GEAR FLOOR BEAM TRUNNION LUG, DRAWINGS 4F23031, 4F23032, 4F23033 AND 4F23034.

6-853. DESCRIPTION. (See figure 6-158-5/2.) The main landing gear floor beams are located in the center fuselage at Fuselage Stations 1303, 1383, 1523, and 1603. Each of the four floor beams has left- and right-side load carrying trunnion lugs at the beam ends in the area of the main landing gear shock strut assembly attachment to the floor beam trunnions. Two bushed holes in the trunnion lugs provide attachment for main landing gear fairing braces. The beams are machined from 7075-T6 aluminum forgings, shot peened, sulfuric acid anodized, and painted with wash primer, epoxy polyamide primer, and two coat of abrasion-resistant polyurethane.

6-854. DEFECTS. Two machined, bushed, 1/2-inch diameter holes in each trunnion lug result in material and grain runout, and interference bushings installed in these holes impose sustained surface tensile stresses on the part. Cracks may originate and propagate radially from the bushed holes in each trunnion lug.

6-855. PRIMARY NDI PROCEDURE - ULTRASONIC.

a. NDI equipment:

(1) Reflectoscope, Sperry UM-715 or equivalent.

(2) Transducer, 5.0 mHz, 0.187-inch, Longitudinal Wave, Sperry P/N 57A2284, or equivalent.

(3) Cable, 6-foot, Microdot/UHF Connector, Sperry P/N 57A2270/or equivalent.

(4) Video Plug-In Module, 10N, Sperry P/N 50E533 or equivalent.

(5) Couplant, Light Oil.

(6) Calibration Standard and Guide, as illustrated in figure 6-158-5/2.

b. Preparation of airplane: No special preparation required.

c. Access: Inspection areas are accessible with the main landing gear down and locked, and outboard main landing gear doors open.

Note

Do not remove bushings from holes in the trunnion lug.

d. Preparation of part: Remove two plates, P/N 4F75926, at Fuselage Stations 1303 and 1603, and two plates, P/N 4F75925, at Fuselage Stations 1383 and 1523 from each trunnion lug in accordance with T.O. 1C-5A-2-2, Section II. Remove only two plates at a time at any one of the four trunnion lugs per side of airplane. See T.O. 1C-5A-23, Section V, for cleaning instructions. Clean local inspection areas as necessary to permit good contact between part and transducer guide.

e. Instrument settings/calibration:

(1) Apply couplant to the calibration standard surface that is opposite the saw cut. Couple the transducer guide to the calibration standard by inserting the pin portion of the guide into the bushed hole in the calibration standard. Insert longitudinal wave transducer into guide hole marked either "CW" or "CCW." Refer to Section I, Initial Shear Wave Calibration - Solid Standard with Contoured Wedge.

(2) Adjust transducer/guide to position No. 1. Calibrate the instrument to display the first interface pulse near the left side of the CRT.

(3) Rotate transducer/guide in a clockwise direction to position No. 2 and note display of saw cut signal on CRT.

(4) Adjust sweep to display the saw cut signal near the right side of the CRT.

(5) Maximize the signal from the saw cut by adjusting the transducer/guide position. Adjust saw cut signal amplitude to 80 percent scope saturation.

(6) Scan inspection surface and observe signal from saw cut. This calibration simulates a crack in the main landing gear floor beam trunnion lug hole radius surface that is common to the bushing outside diameter surface.

f. Inspection: Inspect eight (four each side of airplane) main landing gear floor beam trunnion lugs (two holes per lug) in the hole radius area as illustrated and in accordance with the following:

(1) Apply couplant to the trunnion lug in the area of the bushed hole that is to be covered by the guide. Insert longitudinal wave transducer into the guide hole marked "CW" and couple transducer/guide to the trunnion lug. Insert the guide pin into the lug bushed hole.

(2) Scan the lug in a clockwise direction.

(3) Remove the longitudinal wave transducer from the guide hole marked "CW" and insert into the guide hole marked "CCW." Scan the lug in a counterclockwise direction.

(4) Repeat steps (1) through (3) for the other hole in the trunnion lug.

(5) Repeat steps (1) through (4) to scan the same two holes from the other side of the lug.

6-595

Figure 8. Example of Format and Content of Ultrasonic Inspection Procedures. (Sheet 1 of 6)

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(6) A CRT signal similar to the one displayed for the saw cut in the calibration procedure will indicate a probable crack in the trunnion lug hole radius surface that is common to the bushing outside diameter surface.

g. Mark and report indicated defects. See Section I for approved temporary marking materials.

h. To confirm defect conditions, proceed with Backup NDI Procedure. If no defect is suspected, proceed to System Securing.

6-856. BACKUP NDI PROCEDURE - FLUORESCENT PENETRANT. Perform fluorescent penetrant inspection of suspected defective areas as required.

a. NDI equipment:

(1) Fluorescent penetrant materials and equipment selected from approved list in table 1-III.

b. Preparation of airplane: No special preparation required.

c. Access: Inspection areas are accessible with the main landing gear down and locked, and main landing gear doors open.

d. Preparation of part: With the plates removed from the trunnion lug, remove the bushing from the suspected defective hole if required. Refer to T.O. 1C-5A-3, Section I, for instructions. See T.O. 1C-5A-23, Section V, for finish removal/restoration and cleaning instructions. Strip finish from local suspected defective areas. Clean inspection areas as required to perform fluorescent penetrant inspection (eddy current bolt hole inspection may be used).

e. Instrument settings/calibration: Not applicable.

f. Inspection: Inspect only the suspected defective areas. Perform fluorescent penetrant inspection in accordance with table 1-I, Section I, and instructions contained in T.O. 33B-1-1.

g. Mark and report indicated defects. See Section I for approved temporary marking materials.

6-857. SYSTEM SECURING. Clean areas inspected, restore finishes, reinstall removed components, and perform operational checkouts, as required, in accordance with applicable T.O.'s.

6-596

Figure 8. Example of Format and Content of Ultrasonic Inspection Procedures. (Sheet 2 of 6)

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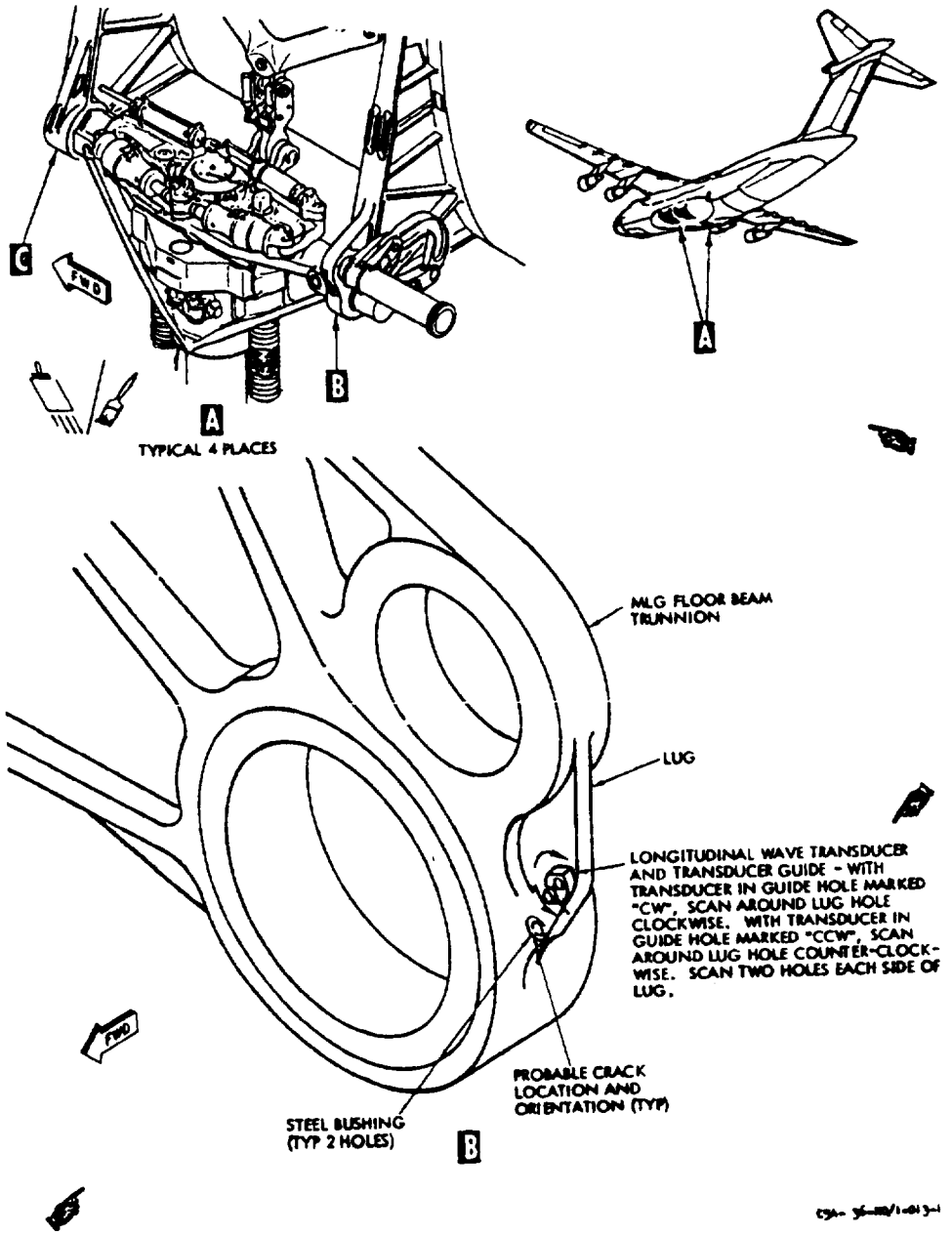


Figure 6-158-5/2. Main Landing Gear Floor Beam Trunnion Lug (Sheet 1 of 4)

6-807

Figure 8. Example of Format and Content of Ultrasonic Inspection Procedures. (Sheet 3 of 6)

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T.O. 1C-5A-36

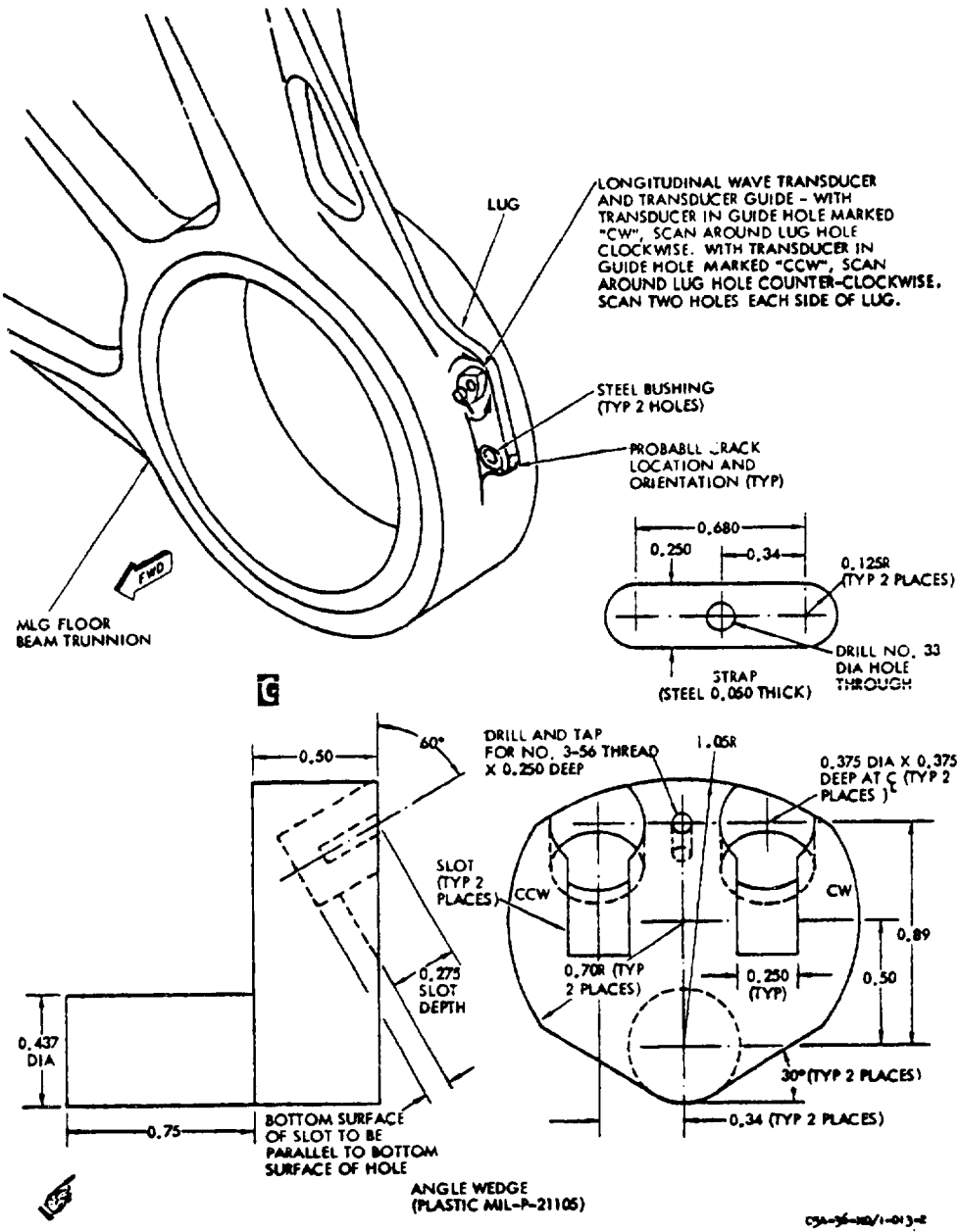


Figure 6-158-5/2. Main Landing Gear Floor Beam Trunnion Lug (Sheet 2 of 4)

6-508

Figure 8. Example of Format and Content of Ultrasonic Inspection Procedures. (Sheet 4 of 6)

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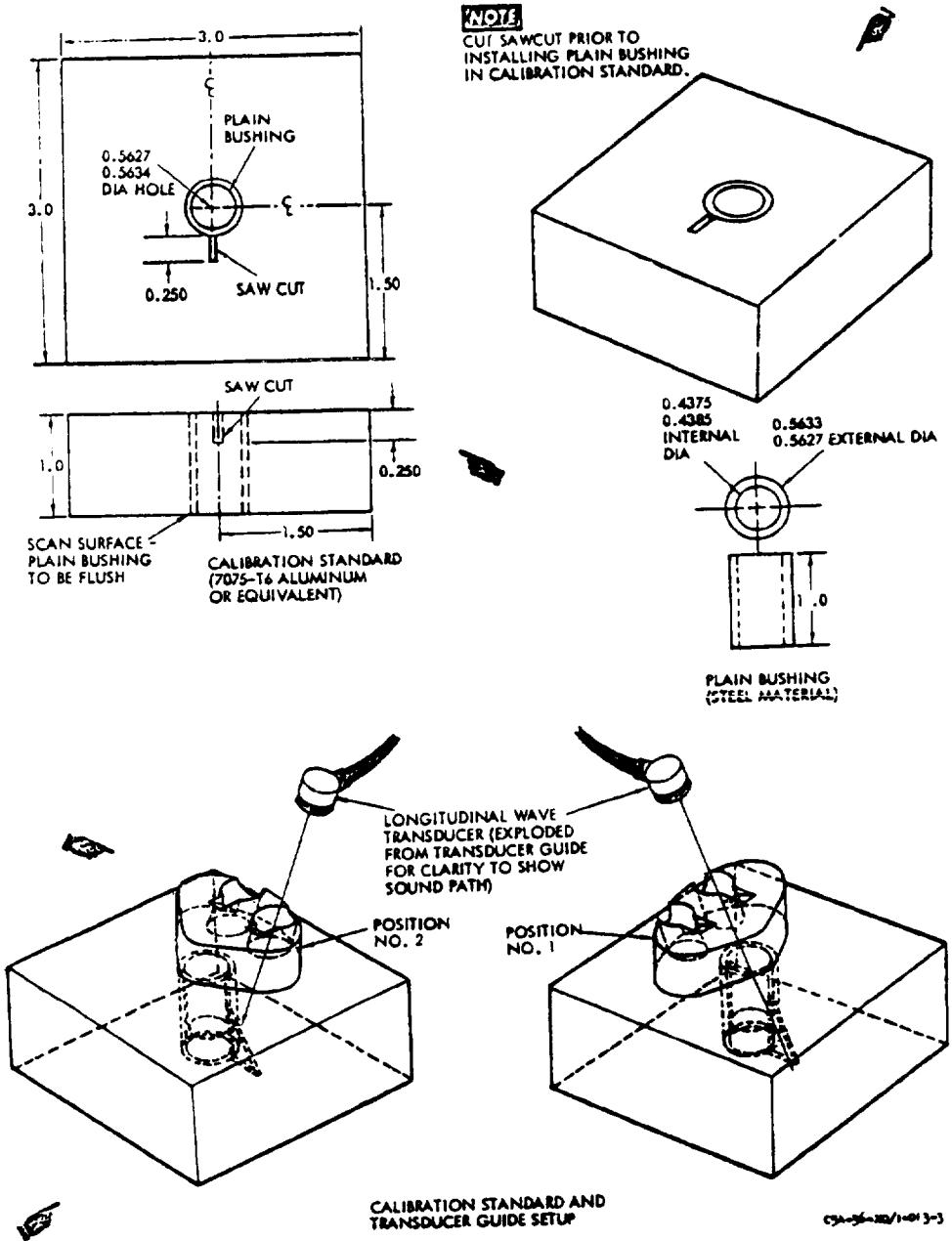


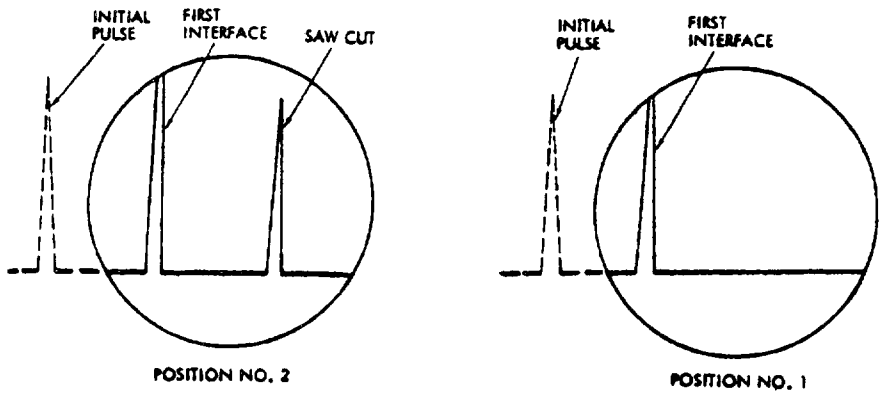
Figure 6-158-5/2. Main Landing Gear Floor Beam Trunnion Lug (Sheet 3 of 4)

8-500

Figure 8. Example of Format and Content of Ultrasonic Inspection Procedures. (Sheet 5 of 6)

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.O. 1C-5A-36



TYPICAL CRT PRESENTATION

Figure 6-158-5/2. Main Landing Gear Floor Beam Trunnion Lug (Sheet 4 of 4)

C5A-36-MD/1-013-A

--800

Figure 8. Example of Format and Content of Ultrasonic Inspection Procedures. (Sheet 6 of 6)

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